

STAT 7L — Statistical Methods for the Biological, Environmental, and Health Sciences Laboratory

Summer Session 2, 2026 · 2 units · In person

Version 1.0 · Revised Jul 22, 2026

Canvas remains authoritative for live instructions, announcements, and deadlines.

Course information	Details
Course dates	July 27 – Aug 28, 2026
STAT 7L parent class	Class number 70059
Instructor	Antonio de Leon
Contact	Email Antonio de Leon · Instructor website
Office hours	By appointment, normally on Zoom; request an appointment through a private Ed Discussion post. Additional drop-in times may be announced in Canvas.
Section 01A	Class 70988 · Monday, 1:00–2:30 p.m. · J Baskin Engineering 156
Section 01B	Class 70989 · Wednesday, 1:00–2:30 p.m. · J Baskin Engineering 156
Concurrent STAT 7 lecture	Class number 70032

Start here. Attend only your officially registered section, bring a charged laptop, open the weekly module in Canvas, and begin assignments early. Monday and Wednesday sections complete equivalent work. Use Ed Discussion for help and a private Ed post for grades, accommodations, group conflicts, enrollment issues, or personal matters.

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Course purpose and learning outcomes

Course description

STAT 7L is the hands-on laboratory companion to STAT 7. You will use R in Google Colab to turn statistical ideas into reproducible analyses of biological, environmental, health, and other applied data. The course emphasizes reasoning, simulation, visualization, inference, modeling, and clear communication—not memorizing software commands. No prior R experience is expected.

STAT 7 and STAT 7L are separate required courses. They have separate Canvas sites, assignments, policies, instructional questions, and grades. Each STAT 7L assignment uses only material covered in STAT 7 by the end of the corresponding Monday lecture. Direct questions about STAT 7 lectures, homework, or examinations to the STAT 7 instructional team and the current STAT 7 Canvas course.

The compressed summer course prioritizes the methods shown in the schedule. One-way ANOVA is an optional, zero-point, ungraded extension outside Gradescope. Power and sample-size calculations are not required.

Learning outcomes

By the end of STAT 7L, you should be able to:

1. Identify observational units, variables, target populations, parameters, study designs, data-quality concerns, missingness, and limits on generalization or causal claims.
2. Produce and interpret numerical summaries and statistical graphics using appropriate variables, units, and comparisons.
3. Calculate and interpret probability, conditional probability, discrete random-variable probabilities, and expected value in context.
4. Use reproducible simulation to investigate randomness, long-run behavior, and sampling variation.
5. Use normal and sampling distributions, standardization, standard error, and the central limit theorem.
6. Apply and interpret confidence intervals and hypothesis tests for one population parameter, including conditions, uncertainty, and contextual conclusions.
7. Select and apply methods using variable types and paired or independent study design, including paired and independent-sample analyses.
8. Assess association between categorical variables with contingency tables and chi-square methods.
9. Fit, diagnose, and interpret a simple linear regression, including predictions, residuals, and limitations.
10. Perform reproducible R analyses in Google Colab and communicate conclusions with evidence, assumptions, uncertainty, units, and appropriate limits.

Materials, technology, and workload

Required materials and services

No textbook is required. All required instructional materials are provided through Canvas. Bring a laptop and charger to every laboratory meeting. A phone or tablet alone may not support the full analysis and submission workflow.

You will use:

- Canvas for instructions, announcements, deadlines, module publication, exit tickets, and grades;

- Google Colab for R notebooks—R and RStudio do not need to be installed locally;
- Google Drive and Google Docs for personal notebook copies and shared group reports;
- Gradescope for group PDF submissions, page mapping, rubrics, and detailed feedback; and
- Ed Discussion as the primary course-help and communication channel.

Use your UCSC account when possible. Confirm that you can access every service before the first graded assignment. If a link, document, software tool, or workflow is inaccessible, report the barrier promptly through a private Ed post so an alternative can be arranged.

Expected workload

UCSC's academic-credit standard estimates 30 total hours of work per unit for the median student. A two-unit course therefore represents approximately 60 hours across this five-week session, or about 12 hours per week. This is an estimate; the time needed will vary by student and assignment.

A typical week may include:

- 1.5 hours in the in-person laboratory;
- 1–2 hours reviewing relevant STAT 7 concepts and preparing;
- 4–6 hours completing notebook analysis and group-report work;
- about 15 minutes completing the individual exit ticket; and
- remaining time reading instructions, coordinating, troubleshooting, revising, exporting, submitting, and reviewing feedback.

The accelerated five-week format moves quickly. Begin early enough to ask questions and resolve access or group problems before a deadline.

Weekly workflow

Unless a module says otherwise, complete these steps each week:

1. Open the weekly Canvas module and review its overview and learning objectives.
2. Save a personal copy of the R Google Colab notebook.
3. Work through the analysis in a group of two or three.
4. Create one shared Google Doc report for the group.
5. Include concise code, relevant output, directly inserted plots, and contextual interpretations.
6. Export one accessible PDF with selectable text.
7. Have one member upload the group PDF to Gradescope.
8. Add every group member and map every question to every relevant page.
9. Reopen the submission and verify that Gradescope received the intended PDF.
10. Complete the individual Canvas exit ticket during your section's window.
11. Review Gradescope feedback and the answer-safe review notes after both sections submit.

Solutions, answer keys, and instructor rubrics are private. Review notes are released only after both sections have submitted.

Assessment and grading

Approved grading weights

Lab 5 Cumulative Group Report is due Friday, **Aug 28, 2026** at 11:59 p.m. PDT. It is the cumulative final evaluation for STAT 7L, is worth 12.5% of the final course grade, and replaces no other listed assignment. There is no separate Saturday examination.

Assessment	Raw points	Final-grade weight	Purpose
Syllabus Acknowledgement	1	0%	Practice and orientation
Lab 0: PDF Submission Practice	1	0%	Practice and orientation
Labs 1–4 Group Reports	75 each	67.5% total	Weekly applied analyses and communication
Lab 5 Cumulative Group Report	100	12.5%	Cumulative method selection, inference, association, and modeling
Lab 1–5 Exit Tickets	10 each	20% total	Individual understanding; lowest score dropped
Total	—	100%	—

Raw points organize grading within each weighted Canvas category or assignment group; they are not added across categories as one unweighted point total. With equal 75-point reports, Labs 1–4 each contribute 16.875% of the final grade under this structure. After the lowest exit ticket is dropped, the remaining four tickets contribute equally within the 20% exit-ticket category.

Canvas uses four weighted assignment groups: Labs 1–4 Group Reports 67.5%, Lab 5 Cumulative Group Report 12.5%, Individual Exit Tickets 20%, and Practice and Orientation 0%.

Group reports

Labs 1–4 are 75 points each. Lab 5 is a 100-point cumulative report and the course’s cumulative final evaluation. Reports assess the analysis, statistical reasoning, evidence, interpretation, communication, and assignment-specific requirements described in the prompt and Gradescope rubric.

Exit tickets

Each exit ticket has five questions, is worth 10 raw points, and ordinarily takes 5–8 minutes. Complete it individually in one attempt; there is no time limit. Section-specific availability begins after each laboratory. Correct answers remain hidden until both sections finish. The lowest of the five scores is dropped. Ordinary makeups are not offered; approved accommodations and documented emergencies are handled individually.

Letter-grade scale

Canvas will display the current course percentage. No automatic rounding is promised.

Grade	Percentage	Grade	Percentage
A+	97–100	C+	77–<80
A	93–<97	C	73–<77
A–	90–<93	C–	70–<73
B+	87–<90	D+	67–<70

Grade	Percentage	Grade	Percentage
B	83–<87	D	63–<67
B–	80–<83	D–	60–<63
		F	<60

Students enrolled Pass/No Pass are evaluated under the [current UCSC grading policy](#), not a separately invented percentage threshold.

Participation, groups, and submissions

Group responsibilities

Groups normally contain two or three students from the same registered section. Submit one shared Google Doc exported as one PDF. Every member's name must appear on the report, and every member must be added to the Gradescope group. All members are responsible for understanding the complete submission, and responsibilities should rotate across analysis, verification, writing, and submission.

Every report includes this contribution statement:

Briefly identify who contributed to analysis, verification, writing, and submission. All group members remain responsible for the complete report.

If a group cannot form, a student may work alone with instructor approval and is not automatically penalized. Raise group conflicts privately before the deadline when possible. Accommodations may require flexibility. A student omitted from a Gradescope group should report the error through a private Ed post within 24 hours.

Attendance and section policy

Attendance at your registered laboratory is expected. Attend only the section in which you are officially enrolled; a section change requires advance instructor approval because room capacity and group coordination are limited. An absence does not automatically turn an otherwise completed report into a zero. An absent student remains responsible for communicating with the group and completing an appropriate share of the report.

Notify Antonio de Leon promptly through a private Ed post about an emergency or approved absence. An alternative participation check may be assigned. DRC accommodations and documented emergencies take precedence. Do not attend another section without permission.

Late-work and extension policy

Each group may request one 24-hour extension for one of Labs 1–4. Submit the request in a private Ed post before the original deadline. The standard extension token does not apply to Lab 5.

Without an approved extension, a group report loses 10% of the assignment's available points per 24-hour period late. Ordinary late submissions are accepted for no more than 48 hours. Work later than that requires an approved accommodation, documented emergency, or explicit instructor authorization. Approved extensions must be reflected in Canvas and Gradescope.

Exit tickets ordinarily have no makeup because the lowest score is dropped. Approved accommodations and documented emergencies are handled individually. Do not disclose personal circumstances in a public post.

Gradescope submission requirements

- One student uploads the group PDF; do not create competing group submissions.
- Add every member before submitting; the maximum group size is three.
- Keep the report's five exact question headings unchanged.
- Map every question to every page containing any part of the response, including continuation pages.
- Use the filename `section_group_labN.pdf`, replacing the section, group, and lab number as directed.
- Reopen the submission and verify the correct PDF. A file existing on one student's computer is not evidence that Gradescope received it.
- Detailed feedback is provided in Gradescope. Course grades appear in Canvas after synchronization and review.

Communication and feedback

Canvas is authoritative for course instructions, deadlines, announcements, module publication, exit tickets, and grades. Ed Discussion is the primary help channel. Search for an existing answer before posting.

Use a regular Ed post for a general question that could help classmates. Use a private Ed post for grades, accommodations, group conflicts, enrollment issues, appointments, or personal matters. Email and Canvas Inbox are reserved mainly for formal administrative use and are generally checked approximately every two days. Duplicate messages across systems do not produce a faster response. Questions sent immediately before a deadline may not receive an immediate answer.

Regrade policy

Submit a regrade request through Gradescope within seven calendar days after a score is released. Identify the specific question and rubric item and explain the perceived grading error. Regrading may include the entire question to apply the rubric consistently; a preference for different partial credit is not by itself evidence of an error. Group members should coordinate to avoid duplicate requests.

Academic integrity, AI, and course materials

Collaboration and individual work

Authorized collaboration is expected on group reports. Every group member must contribute, understand the complete report, and be named accurately. Exit tickets are individual: another person may not answer, revise, or complete one for you.

STAT 7L follows the [UCSC Academic Misconduct Policy for Undergraduates](#). Course-specific prohibited conduct includes submitting another group's work; copying code or prose without authorization; sharing exit-ticket answers before both sections finish; adding a nonparticipant to a submission; fabricating data or output; altering a plot or value to misrepresent results; using unauthorized generative AI; resubmitting prior work without permission; or allowing another person to complete an individual exit ticket. Suspected misconduct is handled through the current university process. This syllabus does not promise a disciplinary sanction outside that process.

Generative-AI policy

Unless an assignment explicitly says otherwise, do not use generative-AI tools to generate, revise, debug, or complete R code, statistical calculations, plots, data analyses, interpretations, written responses, report prose, or exit-ticket answers.

This restriction includes generative assistance built into Google Colab or another coding environment unless an assignment explicitly authorizes its use.

Permitted resources are course materials, R documentation, instructor-approved references, standard spelling and grammar tools that do not generate substantive content, and approved accessibility technologies. You remain responsible for every submitted statement, calculation, plot, and line of code. Unauthorized AI-generated or AI-revised work may be treated as an academic-integrity violation.

If an assignment authorizes a specific AI use, it will state the permitted task, required disclosure, any required citation or transcript, and the limits of assistance. The [UCSC Teaching and Learning Center's AI-use guidance](#) explains why course-specific permission and attribution rules must be explicit.

Intellectual property and privacy

Course materials are provided for enrolled students' educational use. Without written permission, do not upload, sell, redistribute, stream, publish, or post course notebooks, report prompts, rubrics, quiz questions, solutions, Canvas pages, or instructor materials. You retain rights in your original work. Group members may not publicly distribute another student's work without permission. Student-owned working copies remain student-owned; this policy does not transfer them to the instructor.

Accessibility, accommodations, and support

Disability access and digital accessibility

UC Santa Cruz is committed to an accessible learning environment. If you have a disability, or think you may have one, contact the [Disability Resource Center](#) and use [AIM, the DRC portal](#), to request accommodations. If already approved, request your Academic Access Letter and contact the instructor privately so an implementation plan can be arranged. Begin early when possible, but accommodations may be requested at any time. Do not post medical or accommodation information publicly.

Report inaccessible documents, links, software, or workflows promptly through a private Ed post. Alternative formats or workflows will be arranged when needed. You may also use UCSC's [digital accessibility barrier reporting and guidance](#).

Religious accommodations

UCSC provides reasonable religious accommodations without penalty when possible. Contact the instructor privately as early as possible if a class, laboratory, deadline, or assessment conflicts with a religious observance or practice. See UCSC's [religious accommodations guidance](#).

Title IX, CARE, and confidential support

UC Santa Cruz is committed to a learning environment free from gender discrimination and sexual harassment. The instructor and most UCSC employees must report disclosures of possible Title IX misconduct to the Title IX Coordinator. You control whether to respond to outreach and whether to pursue a formal complaint. [CARE](#) provides confidential advocacy, and [Counseling and Psychological](#)

[Services](#) provides confidential counseling. You may also review the [Title IX/CARE syllabus disclosure and reporting resources](#).

Academic, basic-needs, and emergency support

- [Learning Support Services](#) provides peer learning and tutoring support.
- [Slug Support and basic-needs resources](#) can help with food, housing, financial, technology, and other challenges.
- [Student health and counseling services](#) provide health and wellbeing support.
- For urgent campus help, use UCSC's [student urgent-help directory](#). Call 911 for an immediate emergency.

Asking for help early is encouraged. Private circumstances should be shared only through an appropriate private channel.

Five-week schedule

All report deadlines and exit-ticket closing times are 11:59 p.m. Pacific Daylight Time. Canvas remains authoritative; verified changes will be announced there.

Period	Meetings and lecture alignment	STAT 7L work	Exit ticket due	Group report due
Orientation	Module 00 and Lab 0 released Thu Jul 23	Start Here; syllabus review; Colab, Google Docs, PDF, and submission practice	None	Syllabus Acknowledgement and Lab 0 practice: Thu Jul 30
Week 1	01A Mon Jul 27 ; 01B Wed Jul 29	Lab 1: Data, Study Design, and Descriptive Statistics	01A Tue Jul 28 ; 01B Thu Jul 30	Sun Aug 2
Week 2	01A Mon Aug 3 ; 01B Wed Aug 5	Lab 2: Probability and Discrete Random Variables	01A Tue Aug 4 ; 01B Thu Aug 6	Sun Aug 9
Week 3	01A Mon Aug 10 ; 01B Wed Aug 12	Lab 3: Normal and Sampling Distributions; Wednesday follows STAT 7 Exam 1, with no added Wednesday-only content	01A Tue Aug 11 ; 01B Thu Aug 13	Sun Aug 16
Week 4	01A Mon Aug 17 ; 01B Wed Aug 19	Lab 4: Confidence Intervals and One-Parameter Tests	01A Tue Aug 18 ; 01B Thu Aug 20	Sun Aug 23

Period	Meetings and lecture alignment	STAT 7L work	Exit ticket due	Group report due
Week 5	01A Mon Aug 24 ; 01B Wed Aug 26	Lab 5: Multiple Samples, Association, and Modeling; cumulative; Wednesday follows STAT 7 Exam 2, with no added Wednesday-only content	01A Tue Aug 25 ; 01B Thu Aug 27	Lab 5 Cumulative Group Report: Fri Aug 28

One-way ANOVA is optional, worth zero points, ungraded, and outside Gradescope. Power and sample-size calculations are not required. Review notes appear only after both sections have submitted.

Exact exit-ticket windows

Lab	Section 01A window	Section 01B window
Lab 1	Opens Jul 27 at 2:30 p.m.; due Jul 28 at 11:59 p.m.	Opens Jul 29 at 2:30 p.m.; due Jul 30 at 11:59 p.m.
Lab 2	Opens Aug 3 at 2:30 p.m.; due Aug 4 at 11:59 p.m.	Opens Aug 5 at 2:30 p.m.; due Aug 6 at 11:59 p.m.
Lab 3	Opens Aug 10 at 2:30 p.m.; due Aug 11 at 11:59 p.m.	Opens Aug 12 at 2:30 p.m.; due Aug 13 at 11:59 p.m.
Lab 4	Opens Aug 17 at 2:30 p.m.; due Aug 18 at 11:59 p.m.	Opens Aug 19 at 2:30 p.m.; due Aug 20 at 11:59 p.m.
Lab 5	Opens Aug 24 at 2:30 p.m.; due Aug 25 at 11:59 p.m.	Opens Aug 26 at 2:30 p.m.; due Aug 27 at 11:59 p.m.

Course changes and acknowledgement

Course details may be adjusted for instructional, accessibility, safety, university-policy, or scheduling reasons. Substantive changes will be announced through Canvas, and the syllabus revision date will be updated. Review Canvas announcements. Changes will not be applied retroactively in a way that unfairly disadvantages students when avoidable.

Complete the one-point Syllabus Acknowledgement in Canvas. It has unlimited attempts, retains the highest score, belongs to Practice and Orientation, and does not affect the final grade.

Document status: Version 1.0 · Revised Jul 22, 2026